breaks in the clouds of Venus, any more than in the clouds of Titan.* These worlds are permanently overcast.

In the ultraviolet there is detail, but due to transient patches of high-altitude overcast, far above the main cloud deck. The high clouds race around the planet much faster than the planet itself turns: super-rotation. We have an even smaller chance of seeing the surface in the ultraviolet.

When it became clear that the atmosphere of Venus was much thicker than the air on Earth—as we now know, the pressure at the surface is ninety times what it is here—it immediately followed that in ordinary visible light we could not possibly see the surface, even if there *were* breaks in the clouds. What little sunlight is able to make its tortuous way through the dense atmosphere to the surface would be reflected back, all right; but the photons would be so jumbled by repeated scattering off molecules in the lower air that no image of surface features could be retained. It would be like a "whiteout" in a polar snowstorm. However, this effect, intense Rayleigh scattering, declines rapidly with increasing wavelength; in the near-infrared, it was easy to calculate, you *could* see the surface if there were breaks in the clouds—or if the clouds were transparent there.

So in 1970 Jim Pollack, Dave Morrison, and I went to the McDonald Observatory of the University of Texas to try to observe Venus in the near-infrared. We "hypersensitized" our emulsions; the good old-fashioned* glass photographic plates were treated with ammonia, and sometimes heated or briefly illuminated, before being exposed at the telescope to light from Venus. For a time the cellars of McDonald Observatory reeked of ammonia. We took many pictures. None showed any detail. We concluded that either we hadn't gone far enough into the infrared, or the clouds of Venus were opaque and unbroken in the near infrared.

More than 20 years later, the *Galileo* spacecraft, making a close flyby of Venus, examined it with higher resolution and sensitivity, and at wavelengths a little further into the infrared than we were able to reach with our crude glass emulsions. *Galileo* photographed great mountain ranges. We already knew of their existence, though;